

Polynomials

Definitions

A function of the form $f(x) = a_0 + a_1x + \cdots + a_nx^n$ is a *polynomial* in one variable, the numbers $a_0, a_1, \dots, a_n$ are the *coefficients* of the polynomial. Suppose $a_n \neq 0$, then n is the *degree* of f , denoted by $\deg f$. Thus the functions $f(x) = a_0 + a_1x + a_2x^2$ ($a_2 \neq 0$) are polynomials of degree 2 (quadratic functions), the functions $f(x) = ax + b$ ($a \neq 0$) are functions of degree 1 (linear functions), and the functions $f(x) = a$ ($a \neq 0$) are functions of degree 0 (constant functions). By convention $f(x) \equiv 0$ is a function of degree $-\infty$.

Let $f(x) = a_0 + a_1x + \cdots + a_nx^n$, $g(x) = b_0 + b_1x + \cdots + b_mx^m$. Two polynomials are *equal* if and only if all the corresponding coefficients are equal (write $f(x) \equiv g(x)$). Define $f(x) \pm g(x) = (a_0 \pm b_0) + (a_1 \pm b_1)x + \cdots$ (finite terms), i.e., additions and subtractions of polynomials are defined *component-wise*. Also scalar multiplication: $kf(x) = ka_0 + ka_1x + \cdots + ka_nx^n$. Multiplications of polynomials are defined using the *distributive law*: $f(x) \cdot g(x) = a_0b_0 + (a_0b_1 + a_1b_0)x + (a_0b_2 + a_1b_1 + a_2b_0)x^2 + \cdots + a_nb_mx^{n+m}$. The following are easy to check:

1. $f(x) + g(x) = g(x) + f(x)$; $f(x) \cdot g(x) = g(x) \cdot f(x)$ (commutative properties).
2. $f(x) + (g(x) + h(x)) = (f(x) + g(x)) + h(x)$; $f(x)(g(x)h(x)) = (f(x)g(x))h(x)$ (associative properties).
3. $f(x)(g(x) + h(x)) = f(x)g(x) + f(x)h(x)$ (distributive property).
4. If $f(x) \neq 0$, $f(x)g(x) = f(x)h(x)$, then $g(x) = h(x)$ (cancellation property).

Also easy to check:

1. $\deg(f \pm g) \leq \max(\deg f, \deg g)$.
2. $\deg(fg) = \deg f + \deg g$.

Euclidean Algorithm

A basic operation of polynomials is the so-called *Euclidean algorithm*, as it resembles the Euclidean algorithm for integers. To begin, we need the *Division algorithm*: given polynomials $f(x)$ and $g(x)$, then there *exist unique* $q(x)$ (*quotient*) and $r(x)$ ($\deg r(x) < \deg g(x)$) (*remainder*) such that $f(x) = q(x)g(x) + r(x)$. In fact if $\deg f < \deg g$, then we just take $q(x) = 0$ and $r(x) = f(x)$. The more interesting case is of course when $\deg f \geq \deg g$, then we can do *long division*, as we did for integers. Now if $f(x) = q(x)g(x)$, i.e., $r(x) \equiv 0$, then we say $g(x)$ is a *factor* of $f(x)$, (or g *divides* f , written as $g(x) \mid f(x)$, or $f(x)$ is a *multiple* of $g(x)$).

Consider the special case when $g(x) = x - c$ (polynomial of degree 1), then $f(x) = q(x)(x - c) + r$, where r is a constant. Hence $f(c) = r$ (*remainder theorem*). In case $r = 0$, then $f(c) = 0$, i.e. c is a *root* (or *zero*) of $f(x)$, this is so if and only if $x - c$ is a factor of $f(x)$ (*Factor Theorem*). From this we get a polynomial of degree n ($n \geq 1$) has at most n roots. Consequently we have the following *uniqueness theorem*:

Theorem: If $f(x)$ and $g(x)$ are polynomials of degree $\leq n$, and such that $f(x_i) = g(x_i), i = 1, 2, \dots, m$, with $x_1, x_2, \dots, x_m$ distinct numbers and $m > n$. Then $f(x) \equiv g(x)$.

Proof: In fact $f(x) - g(x)$ is a polynomial of degree $\leq n$, but with m distinct roots, $m > n$, by the factor theorem, this is only possible if $f(x) - g(x) \equiv 0$.

Suppose $f(x) = a_0 + a_1x + \dots + a_nx^n$ and $g(x) = x - c$, a simple way of finding the quotient and remainder is by the so-called *synthetic division*. In fact $f(x) = a_nx^n + a_{n-1}x^{n-1} + \dots + a_0 = (x - c)(b_{n-1}x^{n-1} + b_{n-2}x^{n-2} + \dots + b_0) + r$. By comparing coefficients, get

$$a_n = b_{n-1}, a_{n-1} + cb_{n-1} = b_{n-2}, a_{n-2} + cb_{n-2} = b_{n-3}, \dots, a_1 + cb_1 = b_0, a_0 + cb_0 = r.$$

Thus we can write $a_n, a_{n-1}, \dots, a_0$ in a row, copy $a_n = b_{n-1}$, multiply c and add a_{n-1} , get b_{n-2} , and so on. The following example illustrates the algorithm.

Example: Find k so that 3 is a root of $f(x) = x^4 - 5x^3 + 5x^2 + kx + 3$.

We check the remainder by finding $f(3)$ using the synthetic division algorithm:

$$\begin{array}{r|rrrrrr} 1 & -5 & 5 & k & 3 & \\ & 3 & -6 & -3 & 3k-9 & \\ \hline 1 & -2 & -1 & k-3 & 3k-6 & = r \end{array}$$

Thus $f(3) = 3k - 6$, as 3 is a root, $f(3) = 0$, hence $k = 2$.

Example: Find the remainder when dividing $x^{81} + x^{49} + x^{25} + x^9 + x$ by $x^3 - x$.

Write $x^{81} + x^{49} + x^{25} + x^9 + x = x(x^{80} - 1) + x(x^{48} - 1) + x(x^{24} - 1) + x(x^8 - 1) + 5x$, notice that $x^3 - x = x(x^2 - 1)$ divides the first four terms of the expression, hence $5x$ is exactly the remainder. Alternate $x^{81} + x^{49} + x^{25} + x^9 + x = (x^3 - x)Q(x) + R(x)$, where $R(x)$ is of degree ≤ 2 . Put $x = 0, 1, -1$ to get $R(x) = 5x$.

Example: Find the remainder when x^n is divided by $x^2 - x - 1$.

Notice $x^n = Q(x)(x^2 - x - 1) + R(x)$, where $R(x)$ is of degree ≤ 1 . The roots of

$$x^2 - x - 1 = 0 \text{ are } a = \frac{1 + \sqrt{5}}{2} \text{ and } b = \frac{1 - \sqrt{5}}{2}, \text{ when put into the expression, get } R(a) = a^n$$

$$\text{and } R(b) = b^n, \text{ hence } R(x) = R(a) + \frac{R(a) - R(b)}{a - b}(x - a) = \frac{(a^n - b^n)x - ab(a^{n-1} - b^{n-1})}{a - b}.$$

$$\text{Use } a - b = \sqrt{5}, ab = -1 \text{ and } F_n = \frac{a^n - b^n}{\sqrt{5}} \text{ (Fibonacci number) to get } R(x) = F_n x + F_{n-1}$$

$$\text{and } Q(x) = F_1 x^{n-2} + F_2 x^{n-3} + \dots + F_{n-1}.$$

Example: Factorize $P(a, b, c) = (a + b + c)^3 - (a^3 + b^3 + c^3)$.

Regard it as a polynomial of a , then $P(-b, b, c) = 0$, as $a - (-b) = a + b$ is a factor. Likewise $b + c$ and $c + a$ are factors. Thus $P(a, b, c) = K(a + b)(b + c)(c + a)$. Let $a = b = c = 1$, get $K = 3$.

Example: Factorize $P(a,b,c) = a^3 + b^3 + c^3 - 3abc$.

Notice $P(-b-c,b,c) = -(b+c)^3 + b^3 + c^3 + 3(b+c)bc = 0$, hence $a+b+c$ is a factor. Let

$\omega = \frac{-1+i\sqrt{3}}{2}$, ($\omega^3 = 1$ and $\bar{\omega} = \omega^2$), the expression is unchanged if b is replaced by ωb and c by $\omega^2 c$, hence $a + \omega b + \bar{\omega} c$ is a factor. Likewise $a + \bar{\omega} b + \omega c$ is a factor. Together we have $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a + \omega b + \bar{\omega} c)(a + \bar{\omega} b + \omega c)$.

Now suppose $d(x)$ is a polynomial dividing both polynomials $f(x)$ and $g(x)$, then $d(x)$ is a *common factor* of $f(x)$ and $g(x)$. Suppose furthermore for any common factor $h(x)$ of $f(x)$ and $g(x)$, we have $h(x) \mid d(x)$, then $d(x)$ is a *highest common factor (HCF)* or *greatest common divisor (GCD)* of f and g (denoted by $\text{GCD}(f, g)$.) If the GCD of two polynomials is a constant, then f and g are *relatively prime*.

To find a GCD of $f(x)$ and $g(x)$, first suppose $\deg f \geq \deg g$. If $f(x) = q(x)g(x)$, then clearly $g(x)$ is a GCD of f and g . Otherwise $f(x) = q(x)g(x) + r_1(x)$, note that $d(x)$ is a (greatest) common factor of f and g if and only if it is a (greatest) common factor of g and r , hence $\text{gcd}(f, g) = \text{gcd}(g, r_1)$. Apply the Euclidean algorithm again, we have $g(x) = q_1(x)r_1(x) + r_2(x)$, hence $\text{gcd}(f, g) = \text{gcd}(g, r_1) = \text{gcd}(r_1, r_2)$. We note these steps cannot go on indefinitely because the degrees of $r_i(x)$ go strictly down and eventually we must have $r_s(x) = q_{s+1}(x)r_{s+1}(x) + r_{s+2}(x)$, and $r_{s+1}(x) = q_{s+2}(x)r_{s+2}(x)$. This implies $r_{s+2}(x)$ is a GCD of the original polynomials. Also working backward we can see that $r_{s+2}(x)$ is of the form $u(x)f(x) + v(x)g(x)$.

Example: Find a GCD of $f(x) = 4x^4 - 2x^3 - 16x^2 + 5x + 9$ and $g(x) = 2x^3 - x^2 - 5x + 4$.

$$\begin{array}{lcl}
 q_1 = 2 & 0 & \left| \begin{array}{cccccc} f & = & 4 & -2 & -16 & 5 & 9 \\ & = & \underline{4} & \underline{-2} & \underline{-10} & \underline{8} & \\ & & r_1 & = & -6 & -3 & 9 \end{array} \right. & \left| \begin{array}{cccccc} g & = & 2 & -1 & -5 & 4 \\ & = & - & - & \underline{2} & \underline{1} & \underline{-3} & - \\ & & & & -2 & -2 & 4 \end{array} \right. & q_2 = -1 & 1 \\
 q_3 = 2 & 3 & \left| \begin{array}{cccccc} \frac{1}{3}r_1 & = & -2 & -1 & 3 \\ & - & - & \underline{-2} & \underline{2} & - \\ & & & & -3 & 3 \end{array} \right. & \left| \begin{array}{cccccc} & = & = & \underline{-2} & \underline{-1} & \underline{3} \\ & & & r_2 & = & -1 & 1 \end{array} \right. \\
 & & \left| \begin{array}{cccccc} & = & = & & \underline{-3} & \underline{3} \\ & & & & & r_3 = 0 \end{array} \right.
 \end{array}$$

We see that $f(x) = 2x \cdot g(x) + (-6x^2 - 3x + 9)$, $g(x) = \frac{1}{3}(-6x^2 - 3x + 9)(-x+1) + (-x+1)$,

and finally $\frac{1}{3}(-6x^2 - 3x + 9) = (2x+3)(-x+1)$. Hence $-x+1$ is a GCD of f and g .

Working backward, we find $-x+1$ in the form $u(x)f(x) + v(x)g(x)$.

The following pieces of information are useful:

1. $d(x)$ is a common factor of $f(x)$ and $g(x)$. Then $d(x)$ is a GCD of $f(x)$ and $g(x)$ if and only if it is of the form $u(x)f(x) + v(x)g(x)$.

2. If $f(x)$ and $g(x)$ are relatively prime, then there exist $u(x)$ and $v(x)$ such that $u(x)f(x) + v(x)g(x) = a$, where a is a non-zero constant, dividing by a get $\frac{u(x)}{a}f(x) + \frac{v(x)}{a}g(x) = 1$.
3. Another characterization: given non-zero polynomials $f(x)$ and $g(x)$, consider the set of non-zero polynomials of the form $m(x)f(x) + n(x)g(x)$, where $m(x)$ and $n(x)$ are any polynomials. Then among these polynomials, take one of the lowest degree, say $r(x)$, then it can be shown that $r(x)$ divides any polynomial in the set. (Apply the Division algorithm, if $r(x)$ does not divide any polynomial in the set, then the remainder is also in the set and of degree smaller than $r(x)$, contradiction.) From this we see that $r(x)$ divides $f(x)$ and $g(x)$, i.e., it is a *GCD* of f and g .

A polynomial with integer coefficients may not have integer root, it may not even be factorized into polynomials with integer coefficients.

Theorem (Eisenstein): $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ is a polynomial with integer coefficients. Suppose p is a prime such that p does not divide a_n , yet it divides $a_i, 0 \leq i \leq n-1$, and p^2 does not divide a_0 , then $f(x)$ cannot be factorized into polynomials with integer coefficients.

Proof: Suppose $f(x) = (b_m x^m + b_{m-1} x^{m-1} + \dots + b_0)(c_k x^k + c_{k-1} x^{k-1} + \dots + c_0)$. Then $a_0 = b_0 c_0$. As p^2 does not divide a_0 and yet p does, so exactly one of b_0 and c_0 is divisible by p , say c_0 does. Now $b_m c_k = a_n$ is not divisible by p , hence c_k is not a multiple of p . Pick the smallest index r ($r \leq k$) such that c_r is not a multiple of p , i.e. all $c_{r-1}, c_{r-2}, \dots, c_0$ are multiples of p . Then $a_r = b_0 c_r + b_1 c_{r-1} + \dots + b_r c_0 \equiv b_0 c_r \pmod{p}$. Now both b_0 and c_r are not divisible by p , implies $r = n$. Hence the first factor is a constant.

Example: Let p be prime, then $x^p - 1 = (x-1)(x^{p-1} + x^{p-2} + \dots + 1)$. Replace x by $x+1$ to show the second factor cannot be factorized (irreducible).

$$\left(\frac{(x+1)^p - 1}{x} \right) = ((x+1)^{p-1} + (x+1)^{p-2} + \dots + 1), \text{ and consider factorization at the left.}$$

Remarks on Polynomials over $\mathbb{Z}$

It is quite different to consider a polynomial over integers, as compared with over rationals, reals or complex numbers. Indeed the rings $\mathbb{Z}[x]$, as compared with $\mathbb{R}[x]$, or $\mathbb{C}[x]$ are very different, (structure!).

For a polynomial with integer coefficients, i.e., $f(x) \in \mathbb{Z}[x]$, we define divisibility and (ir)reducibility as before. If $f(x) = a_0 + ax + \dots + a_n x^n$, $a_i \in \mathbb{Z}$, then $\text{GCD}(a_0, a_1, \dots, a_n)$ is the *content* of $f(x)$, it is *primitive* if its content is 1. Clearly, any nonzero polynomial can be factored into its content and a primitive polynomial. Also unique factorization holds. Most important of all:

Theorem: In $\mathbb{Z}[x]$, any nonzero polynomial is the product of an integer and a finite number of (first term is positive) primitive polynomials. The factorization is unique up to rearrangement.

Lemma (Gauss): The product of two primitive polynomials is primitive.

Proof: Suppose $f(x) = a_n x^n + \dots + a_0$ and $g(x) = b_m x^m + \dots + b_0$ are primitive but its product is not. So there exists p that divides the coefficients of $f(x)g(x)$, but p does not divide some a_i or b_j , say a_r , r is the smallest index with such property, likewise, b_s , and s the smallest such index. The coefficient of $f(x)g(x)$ is

$$a_r b_s + (a_{r+1} b_{s-1} + \dots) + (a_{r-1} b_{s+1} + \dots),$$

note p divides the second and third group, thus $a_r b_s$, thus a_r or b_s , a contradiction.

Lemma: Suppose $f(x)$ is a nonzero polynomial in $\mathbb{Q}[x]$, then there exists $a \neq 0$, with a rational, and such that $af(x)$ is primitive in $\mathbb{Z}[x]$, also if $bf(x)$ has such a property, then $a = \pm b$.

Proof: Let $f(x) = a_n x^n + \dots + a_0$, where a_i are rational, with $a_n \neq 0$. Choose e such that $ea_0, ea_1, \dots, ea_n$ are integers and $d = \text{GCD}(ea_0, ea_1, \dots, ea_n)$. Then

$\frac{e}{d} f(x) = \frac{e}{d} a_n x^n + \dots + \frac{e}{d} a_0$ is primitive. Also if $g(x) = af(x)$ and $h(x) = bf(x)$, are primitive, then $g(x) = \frac{a}{b} h(x)$, since $g(x)$ is primitive, $\frac{a}{b}$ is an integer and it can only be ± 1 .

(It is clear by clearing denominators and take the GCD of the coefficients.)

Theorem: If $f(x)$ has a non-trivial factorization in $\mathbb{Q}[x]$, so is in $\mathbb{Z}[x]$.

Proof: Indeed $f(x) = g(x)h(x)$, and there exist a, b rationals such that $ag(x), bh(x)$ are primitive, hence $abf(x) = (ag(x))(bh(x))$, but $ag(x)$ and $bh(x)$ are primitive, so is $abg(x)h(x)$, thus $ab = \pm 1$.

Note using these, we can prove the first basic theorem. (Existence by previous analysis, uniqueness by considering the polynomial factorized in $\mathbb{Q}[x]$.) But note $\mathbb{Z}[x]$ is not even an PID (principal domain, check $2, x$, cannot be generated from one polynomial), though it is a UFD (Unique factorization domain) as from the theorem, while $\mathbb{Q}[x], \mathbb{R}[x], \mathbb{C}[x]$ are ED (Euclidean domain).

About factorization, we have Eisenstein. Also we may factor a polynomial.

Theorem: Let $f(x)$ be a nonzero polynomial, if p does not divide the first coefficient, and $f(x)$ cannot be factored mod p , i.e., in $F_p[x]$, then $f(x)$ is irreducible in $\mathbb{Z}[x]$.

Prove by contradiction.

Examples: $x^3 + 2x + 1$ cannot be factored in $F_3[x]$, so is in $\mathbb{Z}[x]$. $x^4 + 1$ cannot be factored in $\mathbb{Z}[x]$, but in $F_2[x]$, $x^4 + 1 = (x + 1)^4$. For $p > 2$,

$$\begin{aligned} f(x) &= x^4 + 1 = (x^2)^2 - (-1) = (x^2 + 1)^2 - 2x^2 \\ &= (x^2 - 1)^2 - (-2)x^2. \end{aligned}$$

But $\left(\frac{-1}{p}\right)\left(\frac{2}{p}\right)\left(\frac{-2}{p}\right) = \left(\frac{4}{p}\right) = 1$, hence at least one of the three terms is 1, i.e., $f(x)$ can be factored.

Example: Let $n > 1$, and $f(x) = a_n x^n + \dots + a_1 x + p \in \mathbb{Z}[x]$, with $|a_1 + a_2 + \dots + a_n| < p$, with p prime, then $f(x)$ is irreducible in $\mathbb{Z}[x]$.

Proof: Suppose x_0 a root in $\mathbb{C}$, and $|x_0| \leq 1$, then

$$p = |a_n x_0^n + \dots + a_1 x_0 + p| \leq |a_n| |x_0|^n + \dots + |a_1| |x_0| \leq |a_n| + \dots + |a_1|,$$

a contradiction. Now suppose $f(x) = g(x)h(x)$, then $p = |f(0)| = |g(0)h(0)|$, impossible, as both $|g(0)|, |h(0)|$ larger than 1 but p is prime. (Note if b is the first coefficient of $g(x)$, then $\frac{g(0)}{b}$ is the product of all roots of $g(x)$, hence $\left|\frac{g(0)}{b}\right| > 1$, as all roots have moduli larger than 1.)

Types of Roots

Rational Roots: Let $f(x) = a_0 + a_1 x + \dots + a_n x^n$, where $a_0, a_1, \dots, a_n$ are integers. Suppose $\frac{p}{q}$ (in lowest term) is a rational root of $f(x)$, then $q|a_n$ and $p|a_0$.

In fact we have $0 = f\left(\frac{p}{q}\right) = a_0 + a_1 \frac{p}{q} + a_2 \frac{p^2}{q^2} + \dots + a_n \frac{p^n}{q^n}$.

Hence $0 = a_0 q^n + a_1 p q^{n-1} + a_2 p^2 q^{n-2} + \dots + a_n p^n$.

Write $a_0 q^n = -(a_1 p q^{n-1} + a_2 p^2 q^{n-2} + \dots + a_n p^n)$, as p divides the right hand side of the expression, it divides the left, but p does not divide q , it must divide a_0 . We get the other condition by writing $a_n p^n$ alone in one side.

Multiple Roots: Suppose $f(x) = (x - c)^k g(x)$, with $k \geq 1$ an integer, and $g(c) \neq 0$, then c is a *multiple root of multiplicity k* of $f(x)$. (In case $k = 1$, we actually have a simple root.) It is not hard to check that c is a multiple root of multiplicity k if and only if $f(c) = f'(c) = \dots = f^{(k-1)}(c) = 0$ and $f^{(k)}(c) \neq 0$.

Real Roots: Let $f(x) = a_0 + a_1x + \cdots + a_nx^n$, where $a_0, a_1, \dots, a_n$ are real numbers, ($f(x)$ a real polynomial). Suppose $r+is$ is a complex root of $f(x)$ with $s \neq 0$, then its conjugate $r-is$ is also a root of $f(x)$.

We may proceed as follows. Write $f(x) = q(x)(x-r-is)(x-r+is) + (Ax+B)$, note that $(x-r-is)(x-r+is) = x^2 - 2rx + (r^2 + s^2)$ is real, hence $q(x)$ and $Ax+B$ are real. Put $x = r+is$ in the equation, get $0 = f(r+is) = A(r+is) + B = (Ar+B) + sAi$. This forces $A=0$ and $B=0$. Now put $x = r-is$ into the equation, get $f(r-is) = A(r-is) + B = (Ar+B) - sAi = 0$, as claimed.

Remark: Likewise if $a+b\sqrt{2}$, say $b \neq 0$, is a root of a polynomial $f(x)$ with integer coefficients, then so is $a-b\sqrt{2}$. etc.

Miscellaneous Remarks: A function $f(x)$ is *odd* if $f(x) = -f(-x)$. Prototypes: $x, x^3, x^5, \dots$ and their linear combinations. Their graphs are symmetric with respect to the origin. It is *even* if $f(x) = f(-x)$. Prototypes: $1, x^2, x^4, \dots$ and their linear combinations. Their graphs are symmetric with respect to the y-axis. Clearly if $f(x)$ is odd, $f(0) = 0$.

We also have the *Descartes' rule of signs*. It asserts that the number of *positive real roots* is at most the number of sign changes in the sequence of polynomial coefficients (omitting the zero coefficients), and that the difference between the numbers is even. This means in particular if the difference is 0 or 1, then so is the number of positive roots. When we multiply the coefficients of odd-power term by -1 , i.e., consider $f(-x)$, and apply the rule, get the number of negative roots. For example, $f(x) = +x^3 + x^2 - x - 1$, there is exactly one sign change, i.e., exactly 1 positive real root. Consider $f(-x) = -x^3 + x^2 + x - 1$, there are two sign changes, thus the original polynomial has 2 or 0 negative roots.

Example: Find all non-constant real polynomials such that $P(x^2) = P(x) \cdot P(x-1)$.

First $P(0) \neq 0$, otherwise $P(1) = P(1)P(1-1) = 0$, $P(4) = P(2)P(2-1) = 0$, $P(25) = P(5)P(5-1) = 0, \dots$, and we get infinitely many roots. Now if a is a root of $P(x)$, then $0 = P(a^2) = P(a)P(a-1)$, iteratively we have $0 = P(a) = P(a^2) = P(a^4) = \dots$, as $P(x)$ is non-constant, some a^i are equal, hence $|a| = 1$. Likewise we have $0 = P((a+1)^2) = P((a+1)^4) = \dots$, hence $|a+1| = 1$. Together we have $(a+1)(\bar{a}+1) = (a+1)(\frac{1}{a}+1) = 1$, get $a^2 + a + 1 = 0$. As $P(x)$ is real, $\bar{a}$ is also a root, hence $(x-a)(x-\bar{a}) = x^2 + x + 1$ is its factor, and there is no other. Also $P(1) \neq 0$, hence $P(0) = 1$. This implies $P(x)$ is of the form $(x^2 + x + 1)^n$, its validity can be readily verified.

Example (USAMO 1976): Given polynomials $P(x), Q(x), R(x)$ and $S(x)$ that satisfy $P(x^5) + xQ(x^5) + x^2R(x^5) = (x^4 + x^3 + x^2 + x + 1)S(x)$, show that $x-1$ is a factor of $P(x)$.

Let $\alpha = e^{\frac{2\pi i}{5}}$, put $\alpha, \alpha^2, \alpha^3$ and α^4 into the identity, and by $\alpha^5 = 1$, get $P(1) + \alpha Q(1) + \alpha^2 R(1) = 0$, $P(1) + \alpha^2 Q(1) + \alpha^4 R(1) = 0$, $P(1) + \alpha^3 Q(1) + \alpha R(1) = 0$, and $P(1) + \alpha^4 Q(1) + \alpha^3 R(1) = 0$. sum to get $4P(1) - Q(1) - R(1) = 0$. Multiplying $P(1) + \alpha Q(1) + \alpha^2 R(1) = 0$ by $\alpha, \alpha^2, \alpha^3, \alpha^4$ and add them, get $-P(1) - Q(1) - R(1) = 0$. Together, get $P(1) = 0$.

Hence any real polynomial may be split into quadratic factors (complex conjugate roots) or linear factors (real roots). Using calculus (continuity argument), we see that if $f(x)$ is an odd degree polynomial, then it has at least one real root. We can also use calculus (maximum and minimum) to estimate the number of real roots of a real polynomial. This also relates to the *partial fraction expansion*. Suppose we encounter a factor of the form $(ax^2 + bx + c)^k$ in the denominator, where $ax^2 + bx + c$ is a real polynomial with a pair of complex roots, then we consider expansion of the form

$$\frac{A_k x + B_k}{(ax^2 + bx + c)^k} + \frac{A_{k-1} x + B_{k-1}}{(ax^2 + bx + c)^{k-1}} + \cdots + \frac{A_1 x + B_1}{(ax^2 + bx + c)^1}.$$

Suppose we encounter a real factor of the form $(rx + s)^l$, then we consider expansion of the form

$$\frac{C_l}{(rx + s)^l} + \frac{C_{l-1}}{(rx + s)^{l-1}} + \cdots + \frac{C_1}{(rx + s)^1}.$$

Interpolations

Example: Find the degree 3 polynomial $P(x)$ such that $P(0) = 0$, $P(1) = P(2) = P(3) = 1$.

Let $P(x)$ be of the form $1 + C(x-1)(x-2)(x-3)$. Put $x = 0$ to get $C = \frac{1}{6}$.

Define $\binom{x}{k} = \frac{x(x-1)(x-2)\cdots(x-k+1)}{k!}$, then we have a degree k polynomial. If x is a positive integer, then we have the *binomial coefficients*.

Example: Given $r \neq 0$, find a polynomial $P(x)$ of degree $\leq n$, and such that $P(j) = r^j$, $j = 0, 1, 2, \dots, n$.

The required polynomial is $\sum_{k=0}^n \binom{x}{k} (r-1)^k$. Its degree is $\leq n$, and

$P(j) = \sum_{k=0}^n \binom{j}{k} (r-1)^k = ((r-1) + 1)^j = r^j$, $j = 0, 1, 2, \dots, n$. By uniqueness property it is the

only polynomial. Notice also $P(n+1) = \sum_{k=0}^n \binom{n+1}{k} (r-1)^k = r^{n+1} - (r-1)^{n+1}$.

Example: (USAMO 1984) Let $P(x)$ be a polynomial of degree $3n$, and such that $P(0) = P(3) = \dots = P(3n) = 2$, $P(1) = P(4) = \dots = P(3n-2) = 1$, and $P(2) = P(5) = \dots = P(3n-1) = 0$. Suppose $P(3n+1) = 730$, find n .

Notice $P(j) - 1 = \frac{2}{\sqrt{3}} \operatorname{Im}(e^{\frac{\pi i}{3}} \omega^j)$, $j = 0, 1, \dots, 3n$, where $\omega = e^{\frac{2\pi i}{3}}$. We try

$P(x) = 1 + \frac{2}{\sqrt{3}} \sum_{k=0}^{3n} \binom{x}{k} \operatorname{Im}(e^{\frac{\pi i}{3}} (\omega - 1)^k)$. It is a polynomial of degree $\leq 3n$. Also for $j = 0, 1, 2, \dots, 3n$,

$$P(j) = 1 + \frac{2}{\sqrt{3}} \sum_{k=0}^{3n} \binom{j}{k} \operatorname{Im}(e^{\frac{\pi i}{3}} (\omega - 1)^k) = 1 + \frac{2}{\sqrt{3}} \operatorname{Im}(e^{\frac{\pi i}{3}} \sum_{k=0}^{3n} \binom{j}{k} (\omega - 1)^k) = 1 + \frac{2}{\sqrt{3}} \operatorname{Im}(e^{\frac{\pi i}{3}} \omega^j).$$

Hence $P(3n+1) = 1 + \frac{2}{\sqrt{3}} \operatorname{Im}(e^{\frac{\pi i}{3}} (\omega^{3n+1} - (\omega - 1)^{3n+1})) = -\frac{2}{\sqrt{3}} \operatorname{Im}(e^{\frac{\pi i}{3}} (\omega - 1)^{3n+1})$. From the

DeMoivre's theorem, get $P(3n+1) = 1 + 2(\sqrt{3})^{3n} \sin \frac{(3n+1)\pi}{6} = 730 = 1 + 3^6$, hence $n = 4$.

We want to find a polynomial $P(x)$ such that $P(a_i) = b_i$, $i = 0, 1, 2, \dots, n$, and the a_i 's are distinct. We first look for $L_k(x)$ such that $L_k(a_j) = 1$ if $k = j$, and $L_k(a_j) = 0$ if $k \neq j$. First $L_k(x)$ is of the form $C(x - a_1) \dots (x - a_{k-1})(x - a_{k+1}) \dots (x - a_n)$, let $x = a_k$, then

$$C = \frac{1}{(a_k - a_1) \dots (a_k - a_{k-1})(a_k - a_{k+1}) \dots (a_k - a_n)}.$$

Now check that $P(x) = \sum_{k=1}^n b_k L_k(x)$. This is the *Lagrange's Interpolation Formula*.

Example: Solve

$$\begin{cases} ax_1 + a^2x_2 + a^3x_3 + a^4x_4 = 1 \\ bx_1 + b^2x_2 + b^3x_3 + b^4x_4 = 1 \\ cx_1 + c^2x_2 + c^3x_3 + c^4x_4 = 1 \\ dx_1 + d^2x_2 + d^3x_3 + d^4x_4 = 1 \end{cases}$$

where a, b, c and d are distinct nonzero numbers.

We transform the problem to a problem of polynomials. We look for a polynomial $P(x)$ of degree ≤ 4 such that $P(0) = 0$, $P(a) = P(b) = P(c) = P(d) = 1$. By the factor theorem

we have $P(x) - 1 = K(x - a)(x - b)(x - c)(x - d)$. Put $x = 0$, get $K = -\frac{1}{abcd}$. By

uniqueness, we have $P(x) = 1 - \frac{(x - a)(x - b)(x - c)(x - d)}{abcd} = x_4x^4 + x_3x^3 + x_2x^2 + x_1x$.

Expand and compare coefficients, get $x_1 = \frac{abc + abd + acd + bcd}{abcd}$,

$$x_2 = -\frac{ab + ac + ad + bc + bd + cd}{abcd}, \quad x_3 = \frac{a + b + c + d}{abcd} \quad \text{and} \quad x_4 = -\frac{1}{abcd}.$$

Example: Solve the system

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \dots a_{2n}x_n = 0 \\ \dots \\ a_{n-1,1}x_1 + a_{n-1,2}x_2 + \dots a_{n-1,n}x_n = 0 \\ a_{n1}x_1 + a_{n2}x_2 + \dots a_{nn}x_n = 1 \end{cases}$$

where $a_{ij} = \frac{1}{i+j-1}, 1 \leq i, j \leq n$.

Let $R(t) = \sum_{k=1}^n \frac{x_k}{t+k}$. Then $R(t) = \frac{P(t)}{Q(t)} = \frac{P(t)}{(t+1)(t+2)\dots(t+n)}$ and $\deg P \leq n-1$. From the system, have $R(0) = R(1) = \dots = R(n-2) = 0$ and $R(n-1) = 1$. Thus $P(0) = P(1) = \dots = P(n-2) = 0$ and $P(n-1) = Q(n-1)$. By the factor theorem $P(x) = Ct(t-1)\dots(t-n+2)$, and $P(n-1) = C(n-1)! = Q(n-1) = n(n+1)\dots(2n-1)$, get $C = (2n-1)\binom{2n-2}{n-1}$, hence $R(t) = (2n-1)\binom{2n-2}{n-1} \frac{t(t-1)\dots(t-n+2)}{(t+1)(t+2)\dots(t+n)}$. Now we need to determine $x_1, x_2, \dots, x_n$ such that

$$(2n-1)\binom{2n-2}{n-1} \frac{t(t-1)\dots(t-n+2)}{(t+1)(t+2)\dots(t+n)} = \frac{x_1}{t+1} + \frac{x_2}{t+2} + \dots + \frac{x_n}{t+n},$$

which can be obtained by partial fractions:

$$x_k = (2n-1)\binom{2n-2}{n-1} \frac{(-k)(-k-1)\dots(-k-n+2)}{(-k+1)\dots(-1)(1)\dots(n-k)}.$$

Miscellaneous Examples

Example: Solve $3x^3 - 7x^2 + 17x - 5 = 0$.

Try all possible rational roots, get $x = \frac{1}{3}$ is a root, hence

$$3x^3 - 7x^2 + 17x - 5 = (3x-1)(x^2 - 2x + 5) = 0. \text{ The roots are } \frac{1}{3}, 1 \pm 2i.$$

Example: Solve $x^4 + 2ax^3 + bx^2 + 2ax + 1 = 0$.

Note $(x^2 + ax + 1)^2 = x^4 + 2ax^3 + (a^2 + 2)x^2 + 2ax + 1$, hence the original equation becomes $(x^2 + ax + 1)^2 - (a^2 + 2 - b)x^2$, which can be factorized by the difference of squares. (Alternatively the equation is a *reciprocal equation* such that the coefficients $a_k = a_{n-k}$, substitute $z = x + x^{-1}$ and reduce it to a lower degree equation.

Example: Define the *Chebyshev polynomials* $T_n(x) = \cos(n \cos^{-1} x)$. Then $T_0(x) = 1$, $T_1(x) = x$ and $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x), n = 1, 2, \dots$. It can be shown that the roots of $T_n(x) = 0$ are $x = \cos(\frac{(2k-1)\pi}{2n}), k = 1, 2, \dots, n$.

Example (IMO 1991 shortlisted): Show that the zeros of $x^8 - 92x^6 + 134x^4 - 28x^2 + 1 = 0$ are $x = \tan \frac{r\pi}{15}, 1 \leq r < 15$ and $\text{GCD}(r, 15) = 1$.

Note that for $r=1,2,4,7,8,11,13,14$, $GCD(r,15)=1$, and if $\theta = \frac{r\pi}{15}$, then $\tan^2 5\theta = 3$ yet $\tan^2 \theta \neq 3$. Let $x = \tan \theta$, then $\tan 5\theta = \frac{\text{Im}(1+ix)^5}{\text{Re}(1+ix)^5} = \frac{x(x^4-10x^2+5)}{5x^4-10x^2+1}$. As $\tan^2 5\theta = 3$, get $\frac{x^2(x^4-10x^2+5)^2}{(5x^4-10x^2+1)^2} = 3$. Simplify to get $x^{10} - 95x^8 + 410x^6 - 430x^4 + 85x^2 - 3 = 0$. Dividing by $x^2 - 3$ to get $x^8 - 92x^6 + 134x^4 - 28x^2 + 1 = 0$.

Example (Cubic equations): By substituting $z = x + \frac{c_1}{3c_0}$ into $c_0x^3 + c_1x^2 + c_2x + c_3 = 0$, one can reduce the equation to $z^3 + pz + q = 0$. Hence it suffices to consider equations of the form $x^3 + px + q = 0$. Let $x = u + v$, then we have $u^3 + v^3 + (u+v)(3uv + p) + q = 0$. Choose u and v so that $uv = -\frac{p}{3}$. Then $u^3v^3 = \frac{-p^3}{27}$ and $u^3 + v^3 = -q$, hence roots of the equation $z^2 + qz - \frac{p^3}{27} = 0$, or $u^3, v^3 = -\frac{q}{2} \pm \sqrt{R}$, with $R = \left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3$. Choose the cube roots $A = \sqrt[3]{-\frac{q}{2} + \sqrt{R}}$, $B = \sqrt[3]{-\frac{q}{2} - \sqrt{R}}$, then $A + B$ is one of the roots, so are $\omega A + \bar{\omega} B$ and $\bar{\omega} A + \omega B$, where $\omega = \frac{-1 + i\sqrt{3}}{2}$ and $\bar{\omega} = \omega^2 = \frac{1}{\omega}$.

Example: Solve $x^3 - 3x + 1 = 0$.

Put $x = u + v, uv = 1$, get $u^6 + u^3 + 1 = 0$. Hence $u^3 = \frac{-1 \pm i\sqrt{3}}{2} = e^{\pm \frac{2\pi i}{3}}$, and hence a solution $A = (e^{\frac{2\pi i}{9}})^{\frac{1}{3}} + B = (e^{-\frac{2\pi i}{9}})^{\frac{1}{3}} = 2\cos \frac{2\pi}{9}$. The other solutions are $\omega A + \bar{\omega} B = 2\cos \frac{4\pi}{9}$ and $\bar{\omega} A + \omega B = 2\cos \frac{6\pi}{9}$.

Example (Quartic equations): For a quartic equation $x^4 + 2ax^3 + bx^2 + 2cx + d = 0$, we have $x^4 + 2ax^3 = -bx^2 - 2cx - d$. Add $2rx^2 + (ax+r)^2$ so that the left hand side becomes a perfect square $(x^2 + ax + r)^2$. The right hand side then becomes $(2r + a^2 - b)x^2 + 2(ar - c)x + (r^2 - d)$. It will be done if it is a perfect square, namely the discriminant $(ar - c)^2 - (2r + a^2 - b)(r^2 - d) = 0$, which can be reduced to $2r^3 - br^2 + 2(ac - d)r + (bd - a^2d - c^2) = 0$, (*resolvent equation*). The equation can always be solved in real r if the coefficients are real.

Vietes' formulae

A function $P(x_1, x_2, \dots, x_n)$ if it is unchanged under any permutation of $x_1, x_2, \dots, x_n$. Let $\sigma_k = \sum x_{i_1} x_{i_2} \dots x_{i_k}$, where the sum is over all $\binom{n}{k}$ possible indices $\{i_1, i_2, \dots, i_k\} \subseteq \{1, 2, \dots, n\}$. σ_k is the k^{th} elementary symmetric function.

Theorem: Every symmetric rational function of $x_1, x_2, \dots, x_n$ is a rational function of $\sigma_1, \sigma_2, \dots, \sigma_n$.

Example: For $n=3$, $\sigma_1 = x_1 + x_2 + x_3$, $\sigma_2 = x_1 x_2 + x_2 x_3 + x_3 x_1$ and $\sigma_3 = x_1 x_2 x_3$. Then $x_1^3 + x_2^3 + x_3^3 = \sigma_1^3 - 3\sigma_1\sigma_2 + 3\sigma_3$ and $\frac{x_2 x_3}{x_1} + \frac{x_3 x_1}{x_2} + \frac{x_1 x_2}{x_3} = \frac{\sigma_1 \sigma_2 - 3\sigma_3}{\sigma_3}$.

The Viète's formulae relate the roots and coefficients of a polynomial. In fact let $f(x) = a_0 + a_1 x + \dots + a_n x^n$ and $\alpha_1, \alpha_2, \dots, \alpha_n$ are roots of the equation. (Here some roots may be equal, a root of multiplicity 2 is counted twice, etc.) Then $f(x) = a_n (x^n + \frac{a_{n-1}}{a_n} x^{n-1} + \dots + \frac{a_0}{a_n}) = a_n (x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_n)$.

By comparing coefficients we get, for instances

$$\begin{aligned} (-1)^n \alpha_1 \alpha_2 \dots \alpha_n &= \frac{a_0}{a_n}, \\ &\dots \\ \alpha_1 \alpha_2 + \alpha_1 \alpha_3 + \dots + \alpha_{n-1} \alpha_n &= \frac{a_{n-2}}{a_n}, \\ -(\alpha_1 + \alpha_2 + \dots + \alpha_n) &= \frac{a_{n-1}}{a_n}. \end{aligned}$$

Hence the elementary symmetric functions are expressed in terms of the coefficients.

Example: Solve the system

$$\begin{cases} x + y + z = 0 \\ x^2 + y^2 + z^2 = 6ab \\ x^3 + y^3 + z^3 = 3(a^3 + b^3) \end{cases}.$$

Define $P(t) = (t-x)(t-y)(t-z) = t^3 + c_1 t^2 + c_2 t + c_3$. Then $c_1 = -(x+y+z) = 0$,

$c_2 = xy + yz + zx = \frac{1}{2}((x+y+z)^2 - (x^2 + y^2 + z^2)) = -3ab$. Finally

$P(x) = P(y) = P(z) = 0$, hence by summing up, get

$(x^3 + y^3 + z^3) + c_1(x^2 + y^2 + z^2) + c_2(x + y + z) + 3c_3 = 0$, or $c_3 = -(a^3 + b^3)$. Thus x, y and z are roots of $t^3 - 3abt - (a^3 + b^3) = 0$. Checking $t = a+b$ is one root, the other two are $\omega a + \bar{\omega} b$ and $\bar{\omega} a + \omega b$.

Theorem (Newton's Identities) Define $S_p = x_1^p + x_2^p + \dots + x_n^p$, where $x_1, x_2, \dots, x_n$ are roots of the equation $x^n + c_1 x^{n-1} + \dots + c_n = 0$, (note how the coefficients are defined). Then

$$\begin{cases} S_1 + c_1 = 0 \\ S_2 + c_1 S_1 + 2c_2 = 0 \\ S_3 + c_1 S_2 + c_2 S_1 + 3c_3 = 0 \\ \vdots \\ S_n + c_1 S_{n-1} + c_2 S_{n-2} + \dots + c_{n-1} S_1 + n c_n = 0 \end{cases}$$

and $S_p + c_1 S_{p-1} + c_2 S_{p-2} + \dots + c_n S_{p-n} = 0$, $p > n$.

(The first part of the identities may be proved by calculus of linear algebra. For the second part, one merely need to substitute the roots into the equation and add them up.)

Example: Given

$$\begin{cases} x + y + z = 1 \\ x^2 + y^2 + z^2 = 2. \\ x^3 + y^3 + z^3 = 3 \end{cases}$$

Calculate $x^4 + y^4 + z^4$.

We have $S_1 = 1, S_2 = 2, S_3 = 3$. Hence $c_1 = -1$, $c_2 = -\frac{1}{2}$ and $c_3 = -\frac{1}{6}$, get

$$S_4 - 3 - 1 - \frac{1}{6} = 0, \text{ or } S_4 = \frac{25}{6}.$$

Example (USAMO 1980): Let $G_n = a^n \sin(nA) + b^n \sin(nB) + c^n \sin(nC)$, where a, b and c are real and $A+B+C$ is a multiple of π . Suppose $G_1 = G_2 = 0$, then $G_n = 0$ for all natural numbers n .

Define $z_1 = a(\cos A + i \sin A)$, $z_2 = b(\cos B + i \sin B)$, $z_3 = c(\cos C + i \sin C)$. As

$G_1 = G_2 = 0$, we have $\text{Im}(z_1 + z_2 + z_3) = \text{Im}(z_1^2 + z_2^2 + z_3^2) = 0$. As $A+B+C$ is a multiple of π , have $\text{Im } z_1 z_2 z_3 = 0$. Check that all elementary functions of z_1, z_2, z_3 ($\sigma_1, \sigma_2, \sigma_3$) are real. Hence $S_n = z_1^n + z_2^n + z_3^n$ are real. Now then $G_n = \text{Im } S_n = 0$.

Cyclotomic Polynomials

Let n be an integer, using all n th primitive roots of unity as roots of the polynomial form the n th cyclotomic polynomial, its degree is $\phi(n)$, its first coefficient is 1,

$$\Phi_n(x) = \prod_{\substack{k=1 \\ (k,n)=1}}^n (x - \zeta^k), \zeta = e^{\frac{2\pi i}{n}}.$$

(Note: $\Phi_n(x)$ is also the monic polynomial of smallest degree with integer coefficients (*minimal polynomial*) with ζ as a root.)

Theorem: $x^n - 1 = \prod_{d|n} \Phi_d(x)$.

“Proof”: This is how all n th roots of unity grouped together.

Theorem: $\Phi_n(x)$ are polynomials with integer coefficients.

(Proof by induction)

Theorem: $\Phi_n(x)$ are irreducible over $\mathbb{Z}[x]$.

Recall *Moebius Inversion Formula*, if $f(n) = \prod_{d|n} g(d)$, then $g(n) = \prod_{d|n} f(d)^{\mu(\frac{n}{d})}$. Hence

we have $\Phi_n(x) = \prod_{d|n} (x^d - 1)^{\mu(n/d)}$. We get $\Phi_1(x) = x - 1$, $\Phi_2(x) = x + 1$,

$\Phi_3(x) = x^2 + x + 1$, $\Phi_4(x) = x^2 + 1$, $\Phi_5(x) = x^4 + x^3 + x^2 + x + 1$, $\Phi_6(x) = x^2 - x + 1$,
 $\Phi_8(x) = x^4 + 1$, $\Phi_9(x) = x^6 + x^3 + 1$, $\Phi_{10}(x) = x^4 - x^3 + x^2 - x + 1$, and $\Phi_{12}(x) = x^4 - x^2 + 1$.
Etc.

Fundamental Theorem of Algebra

We have seen that any real polynomial is a product of quadratic factors and linear factors, thus if complex numbers are allowed to be chosen as roots, then the polynomial has at least one root. In fact it is the case even if the polynomial has complex coefficients. This is the *fundamental theorem of algebra*: A non-zero polynomial with complex coefficients of degree ≥ 1 has at least one root.

Exercises

- Find all polynomials $p(x)$ such that $p(x^2) = p^2(x)$.
- Let $f(x), g(x), h(x)$ be real polynomials such that $f^2(x) = x(g^2(x) + h^2(x))$. Show that $f(x) \equiv g(x) \equiv h(x) \equiv 0$.
- Given non-constant numbers $a_0, a_1, a_2, \dots, a_n$ such that $a_{i-1} + a_{i+1} = 2a_i$ ($i = 1, 2, \dots, n-1$). Show that $a_0 C_0^n (1-x)^n + a_1 C_1^n x(1-x)^{n-1} + \dots + a_{n-1} C_{n-1}^n x^{n-1}(1-x) + a_n C_n^n x^n$ is a degree 1 polynomial.
- Show that $(x-1)^2$ divides $x^{n+1} - n(x-1) - x$.
- Find the remainder when $x^{2007} - 1$ is divided by $(x^2 + 1)(x^2 + x + 1)$.
- Find the remainder when $x^{60} - 1$ is divided by $x^3 - 2$.
- Let $f(x) = x^3 + bx^2 + cx + d$, where b, c and d are integers and $bd + cd$ is odd. Show that $f(x)$ cannot be factorized into two polynomials with integer coefficients.
- Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, where $a_0, a_1, \dots, a_n$ are integers. Suppose there exist an odd integer k and an even integer l such that both $f(k)$ and $f(l)$ are odd, show that $f(x)$ has no integer roots.
- Find all polynomials $p(x)$ that satisfy $xp(x-1) = (x-2)p(x)$.
- Solve the system:

$$\begin{cases} x_1 + x_2 + \dots + x_n = n \\ x_1^2 + x_2^2 + \dots + x_n^2 = n \\ \dots \\ x_1^n + x_2^n + \dots + x_n^n = n \end{cases}$$

- Write $x^4 + x^3 + x^2 + x + 1$ in the form of $f^2(x) - g^2(x)$, where $f(x)$ and $g(x)$ are two polynomials.
- Find n such that $1 + x + \dots + x^{n-1}$ divides $1 + x^2 + x^4 + \dots + x^{2n-2}$.
- Let $f(x) = (x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_n) - 1$, where $\alpha_1, \alpha_2, \dots, \alpha_n$ are integers, show that $f(x)$ is not the product of two non-trivial polynomials with integer coefficients.
- $f(x)$ is a polynomial with integer coefficients, a, b and c are distinct integers such that $|f(a)| = |f(b)| = |f(c)| = 1$. Then $f(x)$ has no integer roots.
- Let $a_i, 1 \leq i \leq 5$, be integers and $f(x)$, a polynomial with integer coefficients be such that $f(a_i) = 5, 1 \leq i \leq 5$. Show that $f(x)$ has no integer roots.
- The polynomial $f(x) = (x-a)(x-b)(x-c)(x-d) - 9$, with a, b, c and d are distinct integers, has integer roots, show that $4 \mid a+b+c+d$.
- (USAMO 1977) Find all positive integers m and n such that $1 + x^n + x^{2n} + \dots + x^{mn}$ is divisible by $1 + x + x^2 + \dots + x^m$.
- (USAMO 1983) Show that the roots of $x^5 + ax^4 + bx^3 + cx^2 + dx + e = 0$ cannot be all real if $2a^2 < 5b$.
- Let $f(x)$ be a polynomial with integer coefficients. Let a, b and c be distinct integers. Show that it is impossible to have $f(a) = b, f(b) = c, f(c) = a$.
- Find all n such that $x^2 - x + 1$ divides $x^{n+1} - x^n + 1$.

21. Factorize $a(b-c)^3 + b(c-a)^3 + c(a-b)^3$.
22. Factorize $8(a+b+c)^3 - (a+b)^3 - (b+c)^3 - (c+a)^3$.
23. Prove that if the polynomial $P(x) = x^n + A_1x^{n-1} + A_2x^{n-2} + \dots + A_{n-1}x + A_n$ assumes integral values for all integral values of x , then it is possible to represent it as a sum of polynomials $P_0(x) = 1, P_1(x) = x, P_2(x) = \frac{x(x-1)}{2}, \dots,$
- $$P_n(x) = \frac{x(x-1)(x-2)\dots(x-n+1)}{1 \cdot 2 \cdot 3 \dots n} = \binom{x}{n},$$
- having the same property and having integral coefficients. Consequently if a polynomial $P(x)$ of degree assume integral values for $n+1$ successive integers x , then it is integral values for all integers x . (Indeed may assume the values are $0, 1, 2, \dots, n$.)
24. (IMO 1993). Show that $f(x) = x^n + 5x^{n-1} + 3, n > 1$, is irreducible in $\mathbb{Z}[x]$.

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1. If $p(x)$ is a constant, then $p(x) = 0, 1$. Otherwise assume $p(x) = a_n x^n + \dots, a_n \neq 0$. By comparing the highest coefficients of $p(x^2)$ and $p^2(x)$, get $a_n = 1$. Compare $p(x^2)$ and $p^2(x)$ again to see $p(x)$ has no other non-constant term. Thus $p(x) = x^n$.
2. Otherwise the degree of the right hand side is odd and the left hand side even.
3. Check that $a_0, a_1, a_2, \dots, a_n$ form an A.P. Substitute every constant in terms of a_0 and the common difference d and simplify.
4. By factor theorem show that $x-1$ is a factor. Use long division to get the quotient, apply the theorem again.
5. Note that $x^{2007} - 1$ is divisible by $x^3 - 1$, thus $x^2 + x + 1$. Hence the remainder is of the form $(x^2 + x + 1)(ax + b)$. Put $x = \pm i$ to find a and b .
6. $2^{20} - 1$, substitute $\sqrt[3]{2}, \omega\sqrt[3]{2}, \bar{\omega}\sqrt[3]{2}$ into the equation $x^{60} - 1 = (x^3 - 2)Q(x) + Ax^2 + Bx + C$, and check.
7. Note that both $b+c$ and d are odd, hence $f(1)$ is odd. assume $x^3 + bx^2 + cx + d = (x+p)(x^2 + qx + r)$, get p odd and $1+p$ even, not possible.
8. First check a_0 is odd, so is $a_0 + a_1 + \dots + a_n$. Then show any $2m$ or $2n+1$ cannot be roots.
9. Check that $x(x-1)$ is a factor, hence $p(x) = x(x-1)q(x)$. Thus $p(x-1) = (x-1)(x-2)q(x-1)$. Hence $q(x) = q(x-1) = q(x-2) = \dots$, or $q(0) = q(1) = q(2) = \dots$, implies $q(x)$ is a constant.
10. Set
$$p(x) = (x-x_1)(x-x_2)\dots(x-x_n)$$
$$= x^n - (x_1 + x_2 + \dots + x_n)x^{n-1} + (x_1x_2 + \dots + x_{n-1}x_n)x^{n-2} - \dots + (-1)^n x_1x_2\dots x_n$$
$$= x^n - \sigma_1 x^{n-1} + \sigma_2 x^{n-2} - \dots + (-1)^n \sigma_n.$$
Multiply the rows of the system by $(-1)^{n-1}\sigma_{n-1}, (-1)^{n-2}\sigma_{n-2}, \dots, -\sigma_1, 1$ respectively, and multiply the equation $1+1+\dots+1=n$ by $(-1)^n\sigma_n$, and add the equations vertically, get $0=np(1)$, hence 1 is a root of $p(x)$, say $x_1 = 1$. Remove x_1 and proceed accordingly, get $x_1 = x_2 = \dots = x_n = 1$.
- 11.

$$\begin{aligned}
& x^4 + x^3 + x^2 + x + 1 \\
&= x^2 \left(x^2 + \frac{1}{x^2} + x + \frac{1}{x} + 1 \right) \\
&= x^2 \left(\left(x + \frac{1}{x} \right)^2 + \left(x + \frac{1}{x} \right) - 1 \right) \\
&= x^2 \left(x + \frac{1}{x} - \alpha \right) \left(x + \frac{1}{x} - \beta \right) \\
&= (x^2 - \alpha x + 1)(x^2 - \beta x + 1) \\
&= (f(x) - g(x))(f(x) + g(x)).
\end{aligned}$$

Solve to get $f(x) = x^2 + \frac{1}{2}x + 1$ and $g(x) = \frac{\sqrt{5}}{2}x$.

12. Note that $(1-x)(1+x+\dots+x^{n-1}) = 1-x^n$ and $(1-x^2)(1+x^2+x^4+\dots+x^{2n-2}) = 1-x^{2n}$. Thus $(1+x)(1+x^2+x^4+\dots+x^{2n-2}) = (1+x+x^2+\dots+x^{n-1})(1+x^n)$. Therefore $1+x+\dots+x^{n-1}$ divides $1+x^2+x^4+\dots+x^{2n-2}$ if and only if $1+x$ divides $1+x^n$, if and only if n is odd.

13. Otherwise $(x-\alpha_1)(x-\alpha_2)\dots(x-\alpha_n)-1 = p(x)q(x)$, then $p(\alpha_i)q(\alpha_i) = -1$, implies $p(\alpha_i) + q(\alpha_i) = 0$, which in terms implies $p(x) + q(x) = 0$ has n integer roots $\alpha_1, \alpha_2, \dots, \alpha_n$. The degrees are not right.

14. If γ is an integer root, then $f(x) = (x-\gamma)g(x)$, hence $1 = |a-\gamma||g(a)| = |b-\gamma||g(b)| = |c-\gamma||g(c)|$. This is not possible since $a-\gamma, b-\gamma, c-\gamma$ are distinct and $g(a), g(b), g(c)$ are integers.

15. Otherwise $f(x) = (x-\beta)g(x)$ and $a_i - \beta, 1 \leq i \leq 5$, are distinct integers that divide 5.

16. If α is an integer root, then $(\alpha-a)(\alpha-b)(\alpha-c)(\alpha-d) \mid 9$, as $\alpha-a, \alpha-b, \alpha-c, \alpha-d$ are distinct, they must be $\pm 1, \pm 3$ (not ± 9), (by the same token). Add to see $a+b+c+d = 4\alpha$, OK.

17. We need $\frac{(x^{n(m+1)}-1)(x-1)}{(x^{m+1}-1)(x^n-1)}$ be a polynomial. As the factors of $x^{n(m+1)}-1$ are distinct, $x^{m+1}-1$ and x^n-1 cannot have common factor other than $x-1$. Thus $m+1$ and n are relatively prime. The condition is also sufficient, clearly.

18. Let $r_i, 1 \leq i \leq 5$, be the roots, then $\sum r_i = -a$ and $\sum r_i r_j = b$. Then $2a^2 - 5b < 0$ if and only if $2(\sum r_i)^2 - 5\sum r_i r_j < 0$ or $\sum (r_i - r_j)^2 < 0$. Hence the result. (Or by calculus, $f'''(x) = 60x^2 + 24ax + 6b$, and its discriminant $288(2a^2 - 5b) < 0$, cannot have real roots, which would imply the original equation cannot have 5 real roots.)

19. Essentially such cyclic property cannot happen. Let $f(a_i) = a_{i+1}$ cyclically (i.e. $f(a_n) = a_1$). Then $a_{i+1} - a_i = f(a_i) - f(a_{i-1}) = (a_i - a_{i-1})Q(a_i, a_{i-1})$, $1 \leq i \leq n$, where the Q 's are integers. By taking the product, get $1 = Q(a_2, a_1)Q(a_3, a_2) \dots Q(a_n, a_{n-1})Q(a_1, a_n)$. Hence $Q(a_{i+1}, a_i) = \pm 1$. Thus $a_{i+1} - a_i = \pm(a_i - a_{i-1})$. Thus $a_1 - a_2 = a_2 - a_3$ or $a_1 - a_2 = -(a_2 - a_3)$. The second one is impossible, otherwise, $a_1 = a_3$. Subsequently $d = a_1 - a_2 = a_2 - a_3 = \dots = a_n - a_1$. Adding up, get $nd = 0$, or $d = 0$.

20. Consider $w^6 = 1$ to see $n \equiv 1 \pmod{6}$.

21. $(a-b)(b-c)(c-a)(a+b-c)$.

22. Let $x = a + b, y = b + c, z = c + a$, reduce the equation to $(x + y + z)^3 - x^3 - y^3 - z^3 = 3(x + y)(y + z)(z + x)$.

23. We need the property: the product of n consecutive integers is divisible by $n!$. This can be proved by counting the number of $p, p^2, \dots$ that appear in the consecutive product and $n!$. Now use induction or otherwise, show that any polynomial $P(x)$ of degree n is a linear combination of $P_0(x), P_1(x), \dots, P_n(x)$, namely

$P(x) = b_n P_n(x) + b_{n-1} P_{n-1}(x) + \dots + b_1 P_1(x) + b_0 P_0(x)$. We now have to show the b_i 's are integers. Now note $P_1(0) = \dots = P_n(0) = P_2(1) = \dots = P_n(1) = P_3(2) = \dots = P_n(2) = \dots = P_{n-1}(n-1) = P_n(n) = 0$, and $P_0(0) = P_1(1) = P_2(2) = \dots = P_{n-1}(n-1) = P_n(n) = 1$. Put 0 in $P(x)$, get $b_0 = P(0)$ an integer. Now $P(1) = b_0 P_0(1) + b_1 P_1(1)$, get $b_1 = P(1) - b_0 P_0(1)$ an integer. Proceed accordingly to get $b_n = P(n) - b_0 P_0(n) - b_1 P_1(n) - \dots - b_{n-1} P_{n-1}(n)$. Hence all $b_0, b_1, \dots, b_n$ are integers. The second part follows using our first observation.

24. (IMO 1993). Show that $f(x) = x^n + 5x^{n-1} + 3, n > 1$, is irreducible in $\mathbb{Z}[x]$.

24. The polynomial has no integer roots. Further considerations to show that it cannot be factored into two non-constant polynomials,

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Extra Exercises (Added July 2021)

1. Let a and b be integers. If $a+b$ is a root of the equation $x^2+ax+b=0$, find the smallest possible value of ab .
2. (Prelim2015) Let $f(x)=x^6-x^5-x^3-x^2-x$ and $g(x)=x^4-x^3-x^2-1$. If a, b, c, d are the four roots to the equation $g(x)=0$, find the value of $f(a)+f(b)+f(c)+f(d)$.
3. Let $f(x)$ be a polynomial of degree 5. If $f(1)=0$, $f(3)=1$, $f(9)=2$, $f(27)=3$, $f(81)=4$ and $f(243)=5$, find the coefficient of x in $f(x)$.
4. Let p, q, r, s be the four roots of the equation $2(10x+13)^2(5x+8)(x+1)=1$. If $pq+rs$ is real, find its value.
5. Determine the number of positive integers m for which there exist nonnegative integers $x_0, x_1, x_2, \dots, x_{2011}$ such that $m^{x_0} = m^{x_1} + m^{x_2} + \dots + m^{x_{2011}}$.

(Minimal Polynomial) Suppose we have an irrational number α and a polynomial with integer coefficients, of smallest possible degree, and with leading coefficient 1, and such that $f(\alpha)=0$. Then $f(x)$ is the minimal polynomial of α over $\mathbb{Q}$. It can be shown, (using division algorithm essentially) suppose $h(x)$ is another polynomial with integer coefficients and such that $h(\alpha)=0$, then $f(x)|h(x)$.

6. Let $f(x)$ be a polynomial with rational coefficients and let α be a real number satisfying $\alpha^3 - 2019\alpha = (f(\alpha))^3 - 2019f(\alpha) = 2021$. Show that $(f^{(n)}(\alpha))^3 - 2019f^{(n)}(\alpha) = 2021$ for any positive integer n .

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1. Let a and b be integers. If $a+b$ is a root of the equation $x^2+ax+b=0$, find the smallest possible value of ab .

1. (Prelim2015) Since $a+b$ satisfies the given equation, we have

$(a+b)^2+a(a+b)+b=0$. Rearranging gives $b^2+(3a+1)b+2a^2=0$. This means b is a root to the equation $x^2+(3a+1)x+2a^2=0$. As a and b are integers, the discriminant $(3a+1)^2-4\cdot 2a^2=a^2+6a+1$ must be a perfect square.

Let $a^2+6a+1=m^2$. Then $(a+3)^2-8=m^2$ and so $(a+3-m)(a+3+m)=8$. As the two terms $a+3-m$ and $a+3+m$ have the same parity (they differ by $2m$ which is even), they can only be 2 and 4, or -2 and -4 (up to permutation). The corresponding possible values of a are 0 and -6 .

When $a=0$, the equation becomes $b^2+b=0$, giving $b=0$ or $b=-1$. When $a=-6$, the equation becomes $b^2-17b+72=0$, giving $b=8$ or $b=9$. The smallest possible value of ab is thus $(-6)\times 9=-54$.

2. (Prelim2015) Let $f(x)=x^6-x^5-x^3-x^2-x$ and $g(x)=x^4-x^3-x^2-1$. If a, b, c, d are the four roots to the equation $g(x)=0$, find the value of $f(a)+f(b)+f(c)+f(d)$.

Since a is a root to $g(x)=0$, we have $a^4-a^3-a^2-1=0$. Using this relation, we get $f(a)=a^6-a^5-a^3-a^2-a=(a^2+1)(a^4-a^3-a^2-1)+a^2-a+1=a^2-a+1$. This also applies to b, c and d , and hence

$$f(a)+f(b)+f(c)+f(d)=(a^2+b^2+c^2+d^2)-(a+b+c+d)+4.$$

Since a, b, c, d are roots to the equation $x^4-x^3-x^2-1=0$, we have $a+b+c+d=1$ and $ab+ac+ad+bc+bd+cd=-1$. It follows that

$$a^2+b^2+c^2+d^2=(a+b+c+d)^2-2(ab+ac+ad+bc+bd+cd)=3$$

and so $f(a)+f(b)+f(c)+f(d)=3-1+4=6$.

3. Let $f(x)$ be a polynomial of degree 5. If $f(1)=0$, $f(3)=1$, $f(9)=2$, $f(27)=3$, $f(81)=4$ and $f(243)=5$, find the coefficient of x in $f(x)$.

3. (Prelim2015) Define $g(x)=f(3x)-f(x)-1$. Then

$g(1)=g(3)=g(9)=g(27)=g(81)=0$. As $g(x)$ is a polynomial with degree at most 5, we have $g(x)=k(x-1)(x-3)(x-9)(x-27)(x-81)$ for some constant k . Since

$g(0)=-1$, we have $k=\frac{1}{1\times 3\times 9\times 27\times 81}$. In other words, we have

$$g(x)=\frac{(x-1)(x-3)(x-9)(x-27)(x-81)}{1\times 3\times 9\times 27\times 81} \text{ and the coefficient of } x \text{ is}$$

$$\frac{1}{81}+\frac{1}{27}+\frac{1}{9}+\frac{1}{3}+\frac{1}{1}.$$

Note that if the coefficient of x in $f(x)$ is c , then the same coefficient in $f(3x)$

would be $3c$ and hence that in $g(x)$ would be $2c$. In other words, the coefficient of x in $f(x)$ is half of that in $g(x)$. Therefore the answer is $\frac{1}{2}\left(\frac{1}{81} + \frac{1}{27} + \frac{1}{9} + \frac{1}{3} + \frac{1}{1}\right) = \frac{121}{162}$.

(Remark: It is very tedious if we try to get the answer by a linear system.)

4. Let p, q, r, s be the four roots of the equation $2(10x+13)^2(5x+8)(x+1)=1$. If $pq+rs$ is real, find its value.

4. (Prelim2010) Let $u=100x^2+260x+160$, the equation becomes $(u+9)u=10$. Hence $u=1$ or -10 . Thus the four roots are $\frac{-13\pm\sqrt{10}}{10}$ and $\frac{-13\pm i}{10}$. Thus the expression is real only when the conjugate multiplied together, get the value $\frac{329}{100}$.

5. Determine the number of positive integers m for which there exist nonnegative integers $x_0, x_1, x_2, \dots, x_{2011}$ such that $m^{x_0} = m^{x_1} + m^{x_2} + \dots + m^{x_{2011}}$.

5. (August test 2011) Let $P(n) = (n^{x_1} + n^{x_2} + \dots + n^{x_{2011}}) - n^{x_0}$. Then $P(m) = 0$ and $P(1) = 2010$. Hence

$(m-1) \mid P(m) - P(1) = -2010$. This implies $m-1$ is a factor of 2010. The number of factors of

$2010 = 2 \times 3 \times 5 \times 67$ is 16 and indeed each factor can be used to construct such a relation.

Let m be such a number. First we take m pieces of 1s ($m^0 + m^0 + \dots + m^0$, whose sum is m).

Then we take $m-1$ pieces of ' m 's ($m^1 + m^1 + \dots + m^1$), and the total sum is $m + (m-1)m = m^2$. Then we take $m-1$ pieces of ' m^2 's ($m^2 + m^2 + \dots + m^2$), and the total sum becomes $m^2 + (m-1)m^2 = m^3$. Then we take $m-1$ pieces of ' m^3 's ($m^3 + m^3 + \dots + m^3$), and so on, until we get exactly 2011 pieces, which is possible since we can find k such that $k(m-1)+1=2011$. We see the sum is a perfect power of m . (For instance, $2^{2010} = (2^0 + 2^0) + 2^1 + 2^2 + \dots + 2^{2009}$,

or

$$3^{1005} = (3^0 + 3^0 + 3^0) + (3^1 + 3^1) + \dots + (3^{1004} + 3^{1004}).)$$

(Minimal Polynomial) Suppose we have an irrational number α and a polynomial with integer coefficients, of smallest possible degree, and with leading coefficient 1, and such that $f(\alpha) = 0$. Then $f(x)$ is the minimal polynomial of α over $\mathbb{Q}$. It can be shown, (using division algorithm essentially) suppose $h(x)$ is another polynomial with integer coefficients and such that $h(\alpha) = 0$, then $f(x) \mid h(x)$.

6. Let $f(x)$ be a polynomial with rational coefficients and let α be a real number satisfying

$\alpha^3 - 2019\alpha = (f(\alpha))^3 - 2019f(\alpha) = 2021$. Show that $(f^{(n)}(\alpha))^3 - 2019f^{(n)}(\alpha) = 2021$ for any positive integer n .

6. (October 2020 test) Indeed check $g(x) = x^3 - 2019x - 2021$ is irreducible over $\mathbb{Z}$, , thus it is the minimal polynomial of α . Now $g(f(\alpha)) = 0$, thus $g(x) \mid g(f(x))$. Replace x by $f(x)$, iterates to get the result.

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